# FIRE PUMP ANNUAL PERFORMANCE TEST

| **Contractor Information** ______ | Test performed for: |
|---|---|
| Name: ______ | |
| Address: ______ | |
| Phone: ______ | |
| Fax: ______ | Test location: |
| Job Name: ______ | |
| Job No.: ______ | |
| Test Date: ______ | |

Test Method:

Test header     Flow meter to drain or suction reservoir     Flowmeter back to suction (closed-loop metering)

| Pump Nameplate Information: ______ | Motor Nameplate Information: ______ |
|---|---|
| Make: ______ | Make: ______ |
| Model: ______ | SN: ______ |
| Stages: ______ | Volts: ______ |
| Coupling: ______ | Amps: ______ |
| SN: ______ | Frame: ______ |
| Imp. Diam.: ______ | Speed: ______ |
| Speed: ______ | HP: ______ |
| Rating Capacity: ______ | |
| Rated Pressure: ______ | |

| Water Supply Type: | Pump Controller Nameplate Information: ______ |
|---|---|
| | Make: ______ |
| | Model: ______ |
| | SN: ______ |
| | Time set for ______ minutes |
| | Turns on at: ______ |
| | Turns off at: ______ |

## FIRE PUMP ANNUAL PERFORMANCE TEST (Continued)

Pressure Maintenance Pump Nameplate Information: ______

Make: ______

Model: ______

SN: ______

Speed: ______

HP: ______

Volts: ______

Pressure Maintenance Pump Controller Nameplate Information: ______

Make: ______

Model: ______

SN: ______

Turns on at: ______

Turns off at: ______

**Flow Test**

| | | |
|---|---|---|
| A. Is a copy of the manufacturer's certified pump test curve attached? | Yes | No |
| B. Test results compared to the manufacturer's certified pump test curve? | Yes | No |
| C. Gauges and other test equipment calibrated? | Yes | No |
| D. No vibrations that could potentially damage any fire pump component? | Yes | No |
| E. The fire pump performed at all conditions without objectionable overheating of any component? | Yes | No |

F. For each test, record the required information for each load condition using the following formulas (or other acceptable methods) and tables:

$$P_{Net} = P_{Discharge} - P_{Suction}$$

$$Q = 29.83\, cd^2 P^{0.5}$$

$$Pv = 0.43352V^2/(2g) = (Q^2)/(890.47D^4)$$

where:

| | |
|---|---|
| $P_{Net}$ | = net pump pressure (psi) |
| $P_{Discharge}$ | = total pressure at the pump discharge (psi) |
| $P_{Suction}$ | = total pressure at the pump suction (psi) |
| $Q$ | = flow through a circular orifice (gpm) |
| $c$ | = nozzle discharge coefficient |
| $d$ | = nozzle orifice diameter (in.) |
| $P$ | = pressure measured on gauge (pitot) |
| $Pv$ | = velocity pressure (psi) |
| $V$ | = velocity of liquid (ft/sec) |
| $g$ | = gravitational constant (32.174 ft/sec) |
| $D$ | = internal pipe diameter (in.) |

## FIRE PUMP ANNUAL PERFORMANCE TEST (Continued)

| Test | Pump speed (rpm) | Suction pressure (psi) | Discharge pressure (psi) | Nozzle size (in.) Nozzle coefficient Pitot readings (psi) | | | | | | Flow (gpm) | Net pressure (psi) | Rpm adjusted net pressure | Rpm adjusted flow (psi) | Suction velocity pressure (psi)[1] | Discharge velocity pressure (psi)[1] | Velocity adjusted pressure (psi)[1] | Oil pressure (psi)[2] | Exhaust back pressure (in. Hg)[2] | Diesel water temperature[2] | Cooling loop pressure (psi)[2] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 | 2 | 3 | 4 | 5 | 6 | | | | | | | | | | | |
| 0% | | | | | | | | | | | | | | | | | | | | |
| 25% | | | | | | | | | | | | | | | | | | | | |
| 50% | | | | | | | | | | | | | | | | | | | | |
| 75% | | | | | | | | | | | | | | | | | | | | |
| 100% | | | | | | | | | | | | | | | | | | | | |
| 125% | | | | | | | | | | | | | | | | | | | | |
| 150% | | | | | | | | | | | | | | | | | | | | |
| 0% | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | |
| 100% | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | |
| 150% | | | | | | | | | | | | | | | | | | | | |

Pump is ☐ constant speed ☐ variable speed

Notes:

[1]Velocity pressure adjustments provide a more accurate analysis in most cases and as a minimum should be included whenever the pump suction and discharge diameters are different and the pump fails by a narrow margin. The actual internal diameter of the pump suction and discharge should be obtained from the manufacturer.

[2]These readings are applicable to diesel engine pumps only. Recording these readings is not specifically required in Chapter 14.

## FIRE PUMP ANNUAL PERFORMANCE TEST (Continued)

For electric motor–driven pumps also record:

| Test | Voltage | | | Amperes | | |
|---|---|---|---|---|---|---|
| | L1-L2 | L2-L3 | L1-L3 | L1 | L2 | L3 |
| 0% | | | | | | |
| 25% | | | | | | |
| 50% | | | | | | |
| 75% | | | | | | |
| 100% | | | | | | |
| 125% | | | | | | |
| 150% | | | | | | |
| 0% | | | | | | |
| | | | | | | |
| 100% | | | | | | |
| | | | | | | |
| 150% | | | | | | |

| | | | |
|---|---|---|---|
| G. For electric motors operating at rated voltage and frequency, is the ampere demand less than or equal to the product of the full load ampere rating times the allowable service factor as stamped on the motor nameplate? | ☐ Yes | No | N/A |
| H. For electric motors operating under varying voltage: | | | |
| 1. Was the product of the actual voltage and current demand less than or equal to the product of the rated full load current times the rated voltage times the allowable service factor? | Yes | No | N/A |
| 2. Was the voltage always less than 5% above the rated voltage during the test? | Yes | No | N/A |
| 3. Was the voltage always less than 10% above the rated voltage during the test? | Yes | No | N/A |
| I. Did engine-driven units operate without any signs of overload or stress? | Yes | No | N/A |
| J. Was the engine overspeed emergency shutdown tested? | Yes | No | N/A |
| K. Was the governor set to properly regulate the engine speed at rated pump speed? | Yes | No | N/A |
| L. Did the gear drive assembly operate without excessive objectionable noise, vibration, or heating? | Yes | No | N/A |
| M. Was the fire pump unit started and brought up to rated speed without interruption under the conditions of a discharge equal to peak load? | Yes | No | N/A |
| N. Did the fire pump performance equal the manufacturer's factory curve within the accuracy limits of the test equipment? | Yes | No | N/A |
| O. Did the electric motor pumps pass phase reversal test on normal and alternate (if provided) power? | Yes | No | N/A |

## FIRE PUMP ANNUAL PERFORMANCE TEST (Continued)

**Multiple Pump Operation**

A. fire pumps are required to operate ☐ in series ☐ in parallel ☐ N/A to meet the maximum fire protection demand.

B. Record the following information for each of the pumps operating simultaneously

| Test | Pump speed (rpm) | Suction pressure (psi) | Discharge pressure (psi) | Nozzle size (in.) Nozzle coefficient / Pitot readings (psi) 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | Net pressure (psi) | Total flow (gpm) | Flow through this pump (gpm) | Rpm adjusted net pressure | RPM adjusted flow (psi) | Suction velocity pressure (psi) | Suction velocity pressure (psi) | Velocity adjusted net pressure (psi) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0% | | | | | | | | | | | | | | | | | | | | | | | |
| 25% | | | | | | | | | | | | | | | | | | | | | | | |
| 50% | | | | | | | | | | | | | | | | | | | | | | | |
| 75% | | | | | | | | | | | | | | | | | | | | | | | |
| 100% | | | | | | | | | | | | | | | | | | | | | | | |
| 125% | | | | | | | | | | | | | | | | | | | | | | | |
| 150% | | | | | | | | | | | | | | | | | | | | | | | |
| 0% | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | | | |
| 100% | | | | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | | | |
| 150% | | | | | | | | | | | | | | | | | | | | | | | |

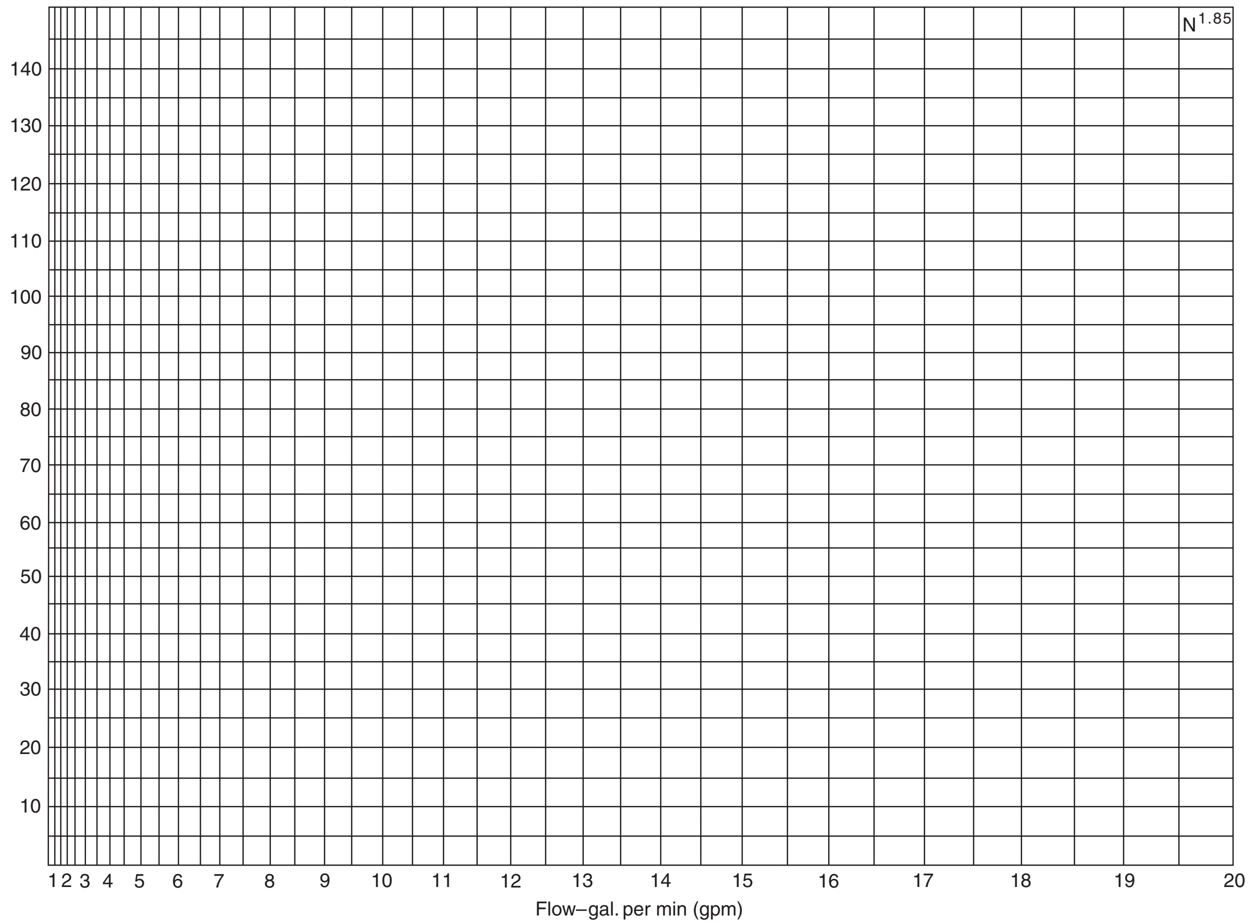
Pump is ☐ constant speed ☐ variable speed

C. Did the fire pump performance equal the manufacturer's factory curve within the accuracy limits of the test equipment during the multiple test? Yes No N/A

| **Additional Tests** | | | **Transfer Switch** |
|---|---|---|---|
| Yes | No | N/A | Simulate power failure while pump is operating at peak load |
| Yes | No | N/A | Power transferred to alternate power source |
| Yes | No | N/A | Pump performed for 2 minutes on alternate power |
| Yes | No | N/A | Record the following while the pump is operating at peak load and alterate power<br>- Voltage where an external means is provided on the controller<br>- Amperage where an external means is provided on the controller<br>- RPM<br>- Suction Pressure psi<br>- Discharge Pressure psi |
| Yes | No | N/A | Pump reconnects to normal power source |
| Yes | No | N/A | Alarm indicating (visual and audible) functions correctly |
| Yes | No | N/A | Suction screens inspected and cleared of any debris |
| Yes | No | N/A | ECM sensors (primary and redundant) function correctly |

## FIRE PUMP ANNUAL PERFORMANCE TEST (Continued)

| | | | |
|---|---|---|---|
| Yes | No | N/A | Pump room temperature control is functioning and being maintained at appropriate temperatures |
| Yes | No | N/A | Parallel and angular alignment of the pump verified |
| Yes | No | N/A | Fire pump supplies 100 percent of the rated flow |
| Yes | No | N/A | Net pressure at each flow point is at least 95% of either the original manufacturers pump curve, original unadjusted field test curve, or test curve generated from the fire pump nameplate. |




**Comments**

Controller: Work shall be completed, to the extent that such work can be completed without opening an energized electric motor-driven fire pump controller.

## POSSIBLE CAUSES OF PUMP TROUBLES

| Fire pump troubles | Suction | | | | Pump | | | | | | | | | | | | | | | Driver and/or Pump | | | | | Driver | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Air drawn into suction connection through leak(s) | Suction connection obstructed | Air pocket in suction pipe | Well collapsed or serious misalignment | Stuffing box too tight or packing improperly installed, worn, defective, too tight, or incorrect type | Water seal or pipe to seal obstructed | Air leak into pump through stuffing boxes | Impeller obstructed | Wearing rings worn | Impeller damaged | Wrong diameter impeller | Actual net head lower than rated | Casing gasket defective, permitting internal leakage (single-stage and multistage pumps) | Pressure gauge is on top of pump casing | Incorrect impeller adjustment (vertical shaft turbine-type pump only) | Impellers locked | Pump is frozen | Pump shaft or shaft sleeve scored, bent, or worn | Pump not primed | Seal ring improperly located in stuffing box, preventing water from entering space to form seal | Excess bearing friction due to lack of lubrication, wear, dirt, rusting, failure, or improper installation | Rotating element binds against stationary element | Pump and driver misaligned | Foundation not rigid | Engine-cooling system obstructed | Faulty driver | Lack of lubrication | Speed too low | Wrong direction of rotation | Speed too high | Rated motor voltage different from line voltage | Faulty electrical circuit, obstructed fuel system, obstructed steam pipe, or dead battery |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| Excessive leakage at stuffing box | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Pump or driver overheats | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Pump unit will not start | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| No water discharge | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Pump is noisy or vibrates | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Too much power required | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Discharge pressure not constant for same gpm | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Pump loses suction after starting | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Insufficient water discharge | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Discharge pressure too low for gpm discharge | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |